

graphically defined rights to slices of the airwaves reinforce the physicality of spectrum. We can't see or touch the radio spectrum, so we envision it as something solid and familiar.

This is a mirage. There is no "aether" over which wireless signals travel; there are only the signals themselves, transmitters and receivers. What we call "spectrum" is simply a convenient way to describe the electromagnetic carrying capacity for the signals. Moreover, the spectrum isn't nearly as congested as we imagine. Run a spectrum analyzer across the range of usable radio frequencies, and the vast majority of what you'll here is silence. Even in bands licensed for popular applications such as cellular telephones and broadcast television, most frequencies are unused most of the time in any given location. This is the case because historical spectrum allocations assume dumb devices that have a hard time distinguishing among signals, thus requiring wide bands with large separation.

With today's technology, the better metaphor for wireless is not land, but oceans. Boats traverse the seas. There is a risk those boats will collide with one another. The oceans, however, are huge relative to the volume of shipping traffic, and the pilots of each boat will maneuver to avoid any impending collision (i.e. ships "look and listen" before setting course). To ensure safe navigation, we have general rules defining shipping lanes, and a combination of laws and etiquette defining how boats should behave relative to one another. A regulatory regime that parceled out the oceans to different companies, so as to facilitate safe shipping, would be overkill. It would sharply reduce the number of boats that could use the seas simultaneously, raising prices in the process.

The same is true with spectrum. Allowing users to share spectrum, subject to rules that ensure they do so efficiently, would be far more effective than turning more spectrum over to private owners.